CS 5326: Advanced Generative AI and Agents

Programming Assignment 2

Instruction Tuning and Direct Preference Optimization on a Small Language Model

Final Report

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1 Repository, Models, and Data

All code, configurations, metrics, generations, pairwise files and judged outputs are in the GitHub repository `https://github.com/AshharShah/small-model-alignment`; the key paths are listed in the final section on limitations and reproducibility.

Models. The SFT model and both from-scratch DPO models are full fine-tunes of Qwen/Qwen2.5-0.5B [1]. Both DPO runs start from the same SFT best checkpoint and use a frozen copy of it as the reference model [2]. The LoRA persona task uses the already instruction-tuned HuggingFaceTB/SmolLM2-360M-Instruct [3], as the assignment specifies.

Training and evaluation data. Table 1 lists the dataset and splits behind each training track, and Table 2 the four fixed evaluation sets. The prompts and their order are fixed, so every model in a comparison is scored on exactly the same inputs.

Table 1. Training data per track. Every dataset was shuffled with seed 42 before filtering.

Track	Source (train / validation split)	Train / val	Filtering
SFT	UltraChat-200k [4] (`train_sft` / `test_sft`)	30,000 / 1,000	first user–assistant turn only; prompt + response + EOS $\leq$ 512 tokens (21,560 of 51,565 train rows dropped)
HH-RLHF DPO	HH-RLHF [5] (`train` / `test`)	15,000 / 1,000	single-turn only (34,382 multi-turn rows dropped); prompt $\leq$ 256 and prompt + response + EOS $\leq$ 512 tokens (127 dropped)
UltraFeedback DPO	UltraFeedback, binarized [6, 7] (`train_prefs` / `test_prefs`)	15,000 / 1,000[†]	single-turn only; chosen score $>$ rejected (3,687 ties dropped); same length limits (12,124 dropped)
LoRA persona DPO	provided `ramsay_style_preferences.json`	198 / 22	none; seeded shuffle (seed 42), 22 rows held out

[†] 27 of the 1,000 validation pairs come from train rows that were not used for training, because the test split has too few usable single-turn pairs.

Table 2. Fixed evaluation sets. "Alignment tax" is the evaluation of a DPO model on the other track's prompts.

Evaluation set	Prompts	Composition	Used for
AlpacaEval-300	300	random subset (seed 42) of the 805 AlpacaEval prompts [8]: self-instruct 95, Koala 60, OASST 54, helpful_base 51, Vicuna 40	SFT vs. DPO-UF (main), DPO-HH (alignment tax)
HH-eval-300	300	synthetic helpful/harmless set provided with the assignment; 10 categories $\times$ 30 prompts	SFT vs. DPO-HH (main), DPO-UF (alignment tax)
Persona-40	40	hand-written, 4 categories $\times$ 10 prompts (cooking, coding, general help, safety-sensitive); disjoint from the training prompts	LoRA adapters vs. base model (test)
Held-out-22	22	validation rows of the persona data, never trained on	LoRA adapter selection (dev)

Seeds, filtering and prompt format. The choices that affect reproducibility are:

- *Seed.* 42 for every shuffle and subset (data order, the AlpacaEval subset, the persona train/validation split), for all training runs, and for the A/B assignment of the pairwise files.
- *Prompt template.* SFT, both DPO runs and all Qwen generations use a single template, `### Instruction:` ⟨instruction⟩ `### Response:`, each part on its own line, with no chat template. The SmolLM2 runs use the model's own chat template for training and generation.
- *Truncation.* An EOS token is appended to every training response before truncation at 512 tokens. All rows were filtered to fit, so no training example is truncated and every response keeps its EOS label.
- *Single-turn only.* Only the first exchange of each UltraChat conversation is used, HH-RLHF and UltraFeedback pairs must be single-turn, and DPO prompts are at most 256 tokens.

References

[1] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[2] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *arXiv preprint arXiv:2305.18290*, 2023.

[3] Loubna Ben Allal et al. SmolLM2: When smol goes big – data-centric training of a small language model. *arXiv preprint arXiv:2502.02737*, 2025.

[4] Ning Ding, Yulin Chen, Bokai Xu, et al. Enhancing chat language models by scaling high-quality instructional conversations. *arXiv preprint arXiv:2305.14233*, 2023.

[5] Yuntao Bai, Andy Jones, Kamal Ndousse, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.

[6] Ganqu Cui, Lifan Yuan, Ning Ding, et al. UltraFeedback: Boosting language models with scaled AI feedback. *arXiv preprint arXiv:2310.01377*, 2023.

[7] Lewis Tunstall, Edward Beeching, Nathan Lambert, et al. Zephyr: Direct distillation of LM alignment. *arXiv preprint arXiv:2310.16944*, 2023.

[8] Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. Length-controlled AlpacaEval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*, 2024.